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New Sizing Rules and Algorithms for Unmanned Lunar Landers

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Abstract

For 37 years between 1976 and 2013, no spacecraft landed softly on the lunar surface. In contrast, since January 2020, nine lunar landers have been launched towards the moon with five landing successfully. The successful spacecraft were Chinese, Indian, Japanese and American.[34] Beyond the Argonaut program, any European space actor interested in cis-lunar exploration will need clear and generalisable sizing rules and sizing algorithms for the effective design of new lunar landers.

This paper presents new sizing rules and sizing algorithms built from a database of fifty lunar landers including thirty-four flight models, and sixteen conceptual designs. The sizing rules are found by statistical relations between the payload mass, the dry mass, the propellant mass, increment of velocity (delta-V), and specific impulse of lunar landers. These statistical rules are found by linear regression and curve fitting techniques. Using these new statistical relations alongside existing parametric subsystem level sizing rules, three iterative sizing algorithms for three propellant types (LOX/LH2, LOX/CH4, and N2O4/AZ) are constructed,[3],[19].[26] They work for payloads from zero to five tons, and they are validated by resizing historical lunar landers and comparing the results. The contributions of this paper are

- The presentation of sizing algorithms which size unmanned lunar landers down to subsystem level using four inputs: (1) the payload mass, (2) the delta-V, (3) specific Impulse and (4) propellant type. The algorithms can be used to size both reusable and non-reusable landers.
- New sizing rules for estimating the dry mass, propellant mass and the subsystem masses of lunar landers.
- The presentation of an open-source and fully cited lunar lander database which includes the mass properties of almost all landers which have been launched towards the moon since 1966 and augmented by a dozen conceptual designs.

Introduction

In the earliest stages of space mission design projects, the designer must be able to consider many different candidate architectures.[51] The information available to the designer about the final definition of the mission is limited. So, in order to compare the performance of each architecture, they are obliged to *estimate* the performance measures of interest to them: for example, the total deliverable payload, the development cost of the spacecraft which carries out the mission, or the cost effectiveness of a reusable vehicle versus an expendable one, or the total launch cost per mission, etc.

The key intermediate parameters for computing mission performance indicators - especially cost related ones - are

1. the payload mass m_p
2. the propellant mass m_{prop}
3. the dry mass m_d
4. the subsystem mass breakdown of the dry mass

An expression which links all of these intermediate parameters is the Tsiolkovsky equation:

$$\Delta V = I_{sp} \cdot g_0 \cdot ln\left(\frac{m_p + m_{prop} + m_d}{m_p + m_d}\right) \tag{1}$$

Where ΔV, is the required change in velocity (m/s), I_{sp} is the specific impulse (m/s) of the propulsion system, g_0 is the gravitational acceleration on Earth.

In order to estimate the key intermediate parameters listed above, the designer can use the Tsiolkovsky equation with some assumptions. The first assumption is for payload mass m_p. This is usually set by the customer at the very earliest stage of a project.[51] The lower bound of ΔV is usually known to within a few percent if the required manoeuvre's are known (in this case descent to the the lunar surface). The choice of specific impulse can be narrowed down to three or four windows around twenty seconds wide depending on the fuel choice. The last parameters which remain are the dry mass m_d, and the propellant mass m_{prop}.

The aim of this paper is to present methods are for estimating these two parameters, and for estimating the subsystem mass breakdown given the minimal information available during the conceptual design phase. Detailed predictions of the mass properties of the spacecraft, will enable more accurate parametric estimation of the life cycle cost of the system.

This paper is divided into five sections. In Section 1, the current state of the art in lunar lander sizing is discussed. In Section 2 the methodology for deriving the sizing rules for lunar landers is described, In Section 3 The sizing rules and algorithms for the subsystem masses are shown and validated against historical lunar lander data. in the lunar lander databases, and in Section 4, the paper is summarised and and the areas of future work are pointed out.

1. State of the Art

Estimating the mass of a proposed spacecraft design is a very important part of the design process because the mass of the spacecraft is the main driver for the life cycle cost of the mission. Parametric Cost estimation relationships which use the mass of a spacecraft to estimate costs have been widely used in industry and academia for many years.[12,23,47] Sometimes, spacecraft mass or initial mass to low Earth orbit are used as proxies for the cost of a mission.[?] This is because more mass in the spacecraft means more mass to launch into space, more material needed to produce the spacecraft, and (usually) more cost in the development of the spacecraft.

This section presents briefly the practice of space systems sizing in general, the application of this practice in early design phases, spacecraft design, followed by the latest work on lunar lander sizing.

1.1 Sizing of aerospace systems

The practice of estimating the size and mass of aerospace systems has been well described in several works. Notably *Spacecraft Systems Engineering*[8] and *Space Mission Analysis and Design*.[51] These two works provide many design rules and sizing methodologies for spacecraft subsystems. The sizing rules discussed in these works can be divided into two types: Bottom-up sizing rules and Top-down sizing rules. For a given subsystem, Bottom-up sizing rules are those which rely on the performance requirement of the system and the system environment for their sizing inputs. For example, given a certain power generation requirement, x and a solar cell technology may have a certain efficiency in turning solar energy into electricity, η, and the solar energy intensity at the spacecraft location, I in W/m^2, the approximate area of the solar array can be estimated by:

$$A = \frac{x}{I \cdot \eta} \tag{2}$$

This is a simple example and Bottom-up sizing rules often require a lot of prior knowledge of the performance requirement of the system or subsystem, the environmental conditions and constraints. Such information may be available in the later stages of design, but it is typically not known in Phase-0 or Phase-A. In this case the designer may use a Top-down sizing rule.

Top-down sizing rules use high level parameters known in the very early stages of the design process such as the payload mass, m_p or the dry mass m_d. For example it may be clear from knowledge of previous spacecraft that the structural mass m_{str} is some fraction, α of the dry mass, m_d.

$$m_{str} = \alpha \cdot m_d \tag{3}$$

These top-down relations require less knowledge about the system being sized and may less accurate than bottom-up relations. Nonetheless, given the right kind of model and enough data, top-down relations can give estimations accurate enough for mission designers conducting trade studies to make decisions about which systems to use or not to use in their mission.

1.2 Sizing of Lunar Landers

For the sizing of lunar landers, a partial consensus has emerged in the literature on the definition of the mass breakdown of lunar landers. There is widespread agreement on the definition of the dry mass, payload mass and the propellant mass but some inconsistency on the definitions of the subsystems worthy of sizing. Table 7 presents a comparison.

Subsystem Mass breakdown conventions					
This Paper	SMAD[51]	SSE[8]	ESAS 2005[1,45]	Ramos 2022[26]	Isaji 2020[19]
Structure	Struct.	Structure	Structure	Structure	Structure + TPS
Avionics	TT&C	TT&C	Avionics	Avionics	Avionics
Power	Power	Power	Power	Power	Power
Thermal	Thermal	Thermal	Protection	Thermal Control	Thermal $\in Structure + TPS$
ADCS $\in Avionics$	ADCS	AOCS	Control (AOCS - Control) $\in Avionics$	AOCS $\in Avionics$	AOCS $\in \{Propulsion, Other\}$
Propulsion	Propul.	Propulsion	Propulsion	Tanks Engines	Propulsion
Other	N/A	Data Handling	Environment Growth Other	N/A	ECLSS Other

Table 1: Comparison of subsystem mass breakdown conventions across different studies.

This table shows on the left column, this paper's subsystem breakdown for the spacecraft of interest, lunar landers. The remaining columns show the equivalent spacecraft subsystems as they appear by different names across the literature. The SMAD and the SSE columns show the subsystem mass breakdown for Earth orbiting and interplanetary unmanned spacecraft. For these spacecraft there is almost complete agreement across the subsystem definitions. This is because their production has been ongoing for many decades, and also because previous editions of these two textbooks themselves have been influential in cementing the consensus on the subsystem definitions for these kinds of spacecraft.

There is less agreement in the design of lunar landers. The remaining three columns, (ESAS, Ramos, and Isaji) show the subsystem definitions used in a series of lunar lander conceptual design studies (ESAS) and sizing methods (Ramos and Isaji). The subsystem mass definition can vary in this field for many reasons: whether the lander is manned or not, what is the validation case for the sizing method etc.

The key takeaway is that when undertaking a sizing study for a kind of spacecraft for which there is only a partial consensus on the subsystem definition, it is important to clearly define and compare this paper's definitions to that of the state of the art. This paper builds upon the work done here by re-using applicable sizing rules for certain subsystems, and developing new ones where the definitions are not compatible.

1.3 Latest Methods

Ramos and Isaji represent the latest two works on lunar lander sizing methodologies. Ramos et al focus on physics based sizing rules for lander subsystem masses, even sizing at the component level in the propulsion subsystem.[26] Ramos also compiles a small database to use statistical sizing rules to size things like the structural mass to dry mass ratio. This physics based approach means that the sizing results are explainable in terms of environmental and mission variables. The sizing rules are collated into a non-iterative sizing routine which was validated on the descent stage of the Altair unmanned lunar lander. The routine sizes the Altair lander dry mass to within 3.5%. However, this was using the known structural mass fraction of 43% instead of the 30% structural mass fraction suggested by the sizing rule they derive from their database. Ramos writes that If their sizing rule had been used for the validation the validation case would have been drastically undersized.

Isaji et al in 2020 relied more on top-down sizing rules based on statistics. They constructed a large database of up to 97 lunar landers, and analysing trends in the database they were able to make multi-variate sizing rules for

Table 2: The definition of each parameter collected on each lunar in the database

Property	definition	units (if any)
year	Launch year	[year]
Organisation	Space agency or Company of origin	
Spacecraft Model	Name of the spacecraft	
Mission name	Name of the Mission	
total mass	$m_t = m_d + m_p + m_{prop}$	[kg]
dry mass	m_d	[kg]
payload mass	m_p	[kg]
prop mass	m_{prop}	[kg]
propellant	The type of fuel and oxidizer	
Specific Impulse	I_{sp}	[s]
Estimated delta-v	ΔV	[m/s]

landers at the subsystem level. The sizing rules were collated together into an iterative sizing algorithm which used dry-mass summation convergence to validate its estimations of the subsystem dry masses. The validation case is the Apollo 17 lunar lander and its dry mass is sized to within 1% of the true value. Unfortunately, the database which made these rules is not directly available so the sizing rules found by Isaji cannot be re-derived.[18] The database includes conceptual lander designs alongside historical flight models which may introduce some errors in the sizing rules when they are validated against historical landers since the conceptual landers are unfinished projects which are not flight tested. Finally since the latest entry in the Isaji database in 2017 there have been 14 additional lunar landers which have been launched towards the moon.

This paper builds upon the latest work in lunar lander sizing methods in three ways:

- Building and publishing an up-to-date database of historical lunar landers.
- Deriving new sizing rules based on this data for unmanned lunar landers.
- Modelling the uncertainty in the mass estimation relationships, and examining its impact on the final sizing.

2. Methodology

This section describes how the lunar lander sizing sizing rules and algorithms were derived and validated using data from historical flight models and subsystem breakdown data from historical flight models and conceptual designs.

2.1 Compiling Databases

The derivation and validation of sizing rules for lunar landers requires data. It is said in aircraft design that simulation data is good, experimental wind tunnel data is better, but that flight data is king. The same can be said for lunar landers. There have been many conceptual lander studies for all payload sizes such as those compiled in the *After LM* catalogue.[1] But the best data comes flight models. This is because during the subsequent project phases after conceptual design, many design changes can happen which can invalidate the sizing relationships used. When data from flight models is used, these design changes are captured in the data. Therefore the sizing rules derived in this paper rely as much as possible on the database of lunar lander flight models shown in section 3.2. This database is an extension of the database used in a previous work by the author. It contains 35 flight models from the fist successful lander, Luna 9 in 1966, up to the latest successfully launched lander, Hakuto-R M2 in 2025.

The data collected on each lander is shown in table 2. The database includes one and two stage landers, including the Apollo lunar modules. The payload mass for two stage landers, manned or unmanned is the total mass of the ascent stage. This maintains comparability between these different lander types.

For the subsystem mass breakdown, data is less readily available. This paper relies heavily on the data in the *After LM* catalogue.[1] The data is shown in table XX.

2.2 Sizing Rule Derivation

To approach the sizing of a spacecraft, a reference model and a reference mission must be defined. For this paper, the reference model is a single stage unmanned lunar lander. Additional stages or manned portions of the spacecraft can

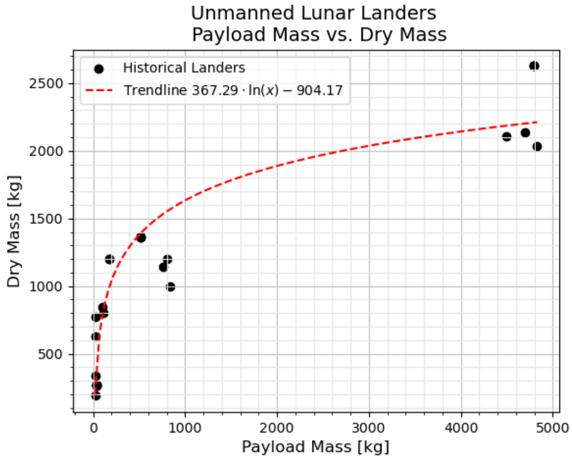

Figure 1: curve of best fit between payload mass and dry mass. $R = 0.92$. Datapoints from table 6

Table 3: Subsystem Definitions

Subsystem	definition
Structure	The hardware in-between the other subsystems which transfers loads between them
Avionics	All electronics sensors and actuators onboard the spacecraft including those for attitude determination and control
Power	The electrical power generation, storage and distribution hardware
Thermal	The active and passive components responsible for the the temperature control of the spacecraft
Propulsion	The tanks, the engines, and propellant management hardware
Other	All other components on board

be modelled as the payload of the lande. The reference mission is a descent and landing from low lunar orbit (LLO) to the lunar surface and (optionally) back again to LLO. Different start and end points of the mission can be modelled by increasing or decreasing the ΔV requirement.

To decide how to use the data to make sizing rules, the sizing algorithms of interest have to be defined. A sizing algorithm is a set of steps to size a system. It is also known as a sizing routine. The algorithms of interest for this paper are those which can size the major mass properties of the system: m_d and m_{prop}, and the subsystem mass breakdown from a given payload mass, m_p , a known destination ie. a known ΔV and a choice of a few fuel types which amounts to a choice of I_{sp}. The subsystems are sized so that their estimated mass can be used to estimate the life cycle cost of the system in future work and so that these algorithms can be compared to the state of the art in section 1.

Figure 3 shows an example of the kind sizing algorithms used in this paper. The common sizing rules required by all the algorithms here are equation 4 and 5. The first is a purely statistical relation to give a first estimate of the dry mass. The second is a re-arrangement of the Tsiolkovsky equation to estimate and the propellant mass with an additional factor β to account for propellant margin.

$$m_d = \alpha \cdot ln(m_p) + \beta \tag{4}$$

This dry mass estimation, is then used to estimate the propellant mass, m_{prop}

$$m_{prop} = \Theta \cdot (m_p + m_d) \cdot (exp(\frac{\Delta V}{I_{sp} \cdot g_0}) - 1) \tag{5}$$

Where Θ the observed fuel margin present in lunar landers in the database. In figure 2, Θ corresponds to the difference in slope between the data trend-line and the prediction line drawn by the Tsiolkovsky equation.

These estimates of the propellant and dry mass are then used alongside the m_p, ΔV and the I_{sp} to estimate the six subsystem masses. Their definitions are shown in table 3. A total of eleven sizing algorithms were investigated for how well they fit the data. The differences between them were how they sized the subsystems and how many of the subsystem mass estimations were fed back to re-estimate the m_{dry}, and the m_{prop}. The sizing rules for the subsystem masses are derived using two different methods single/multiple linear regressions with m_d, m_{prop}, m_p, I_{sp}, and the ΔV see figure 4; and physics-based sizing of components within the subsystem.

The propulsion subsystem can be sized by estimating the size of the major components such as the tanks, the engines, and the propellant management hardware. It is this subsystem which distinguishes the three sizing algorithms

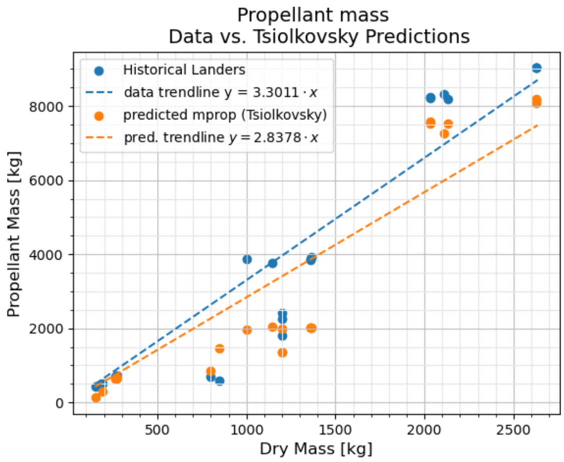

Figure 2: Trendlines for dry mass versus propellant mass of data and propellant mass of prediction

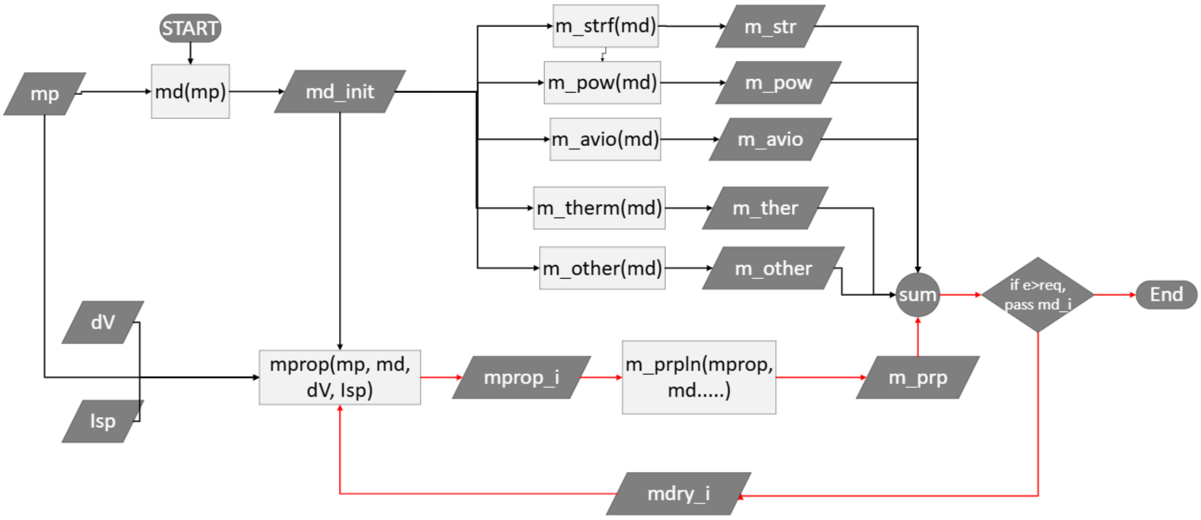

Figure 3: Sizing algorithm format. Iterative portion shown in red. For each fuel type, the propulsion subsystem sizing routine is changed.

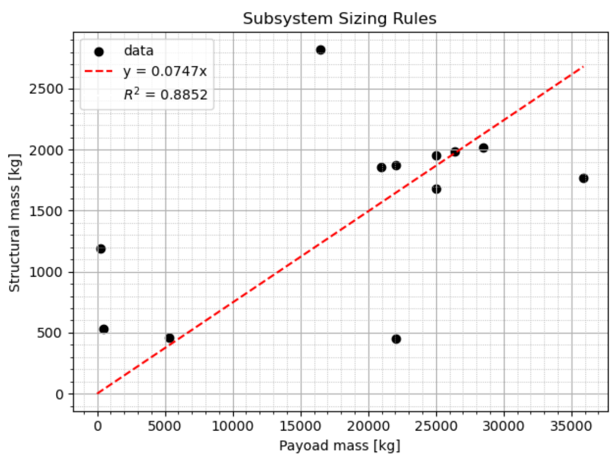

Figure 4: Example of a linear subsystem sizing rule found fro the database in 7.

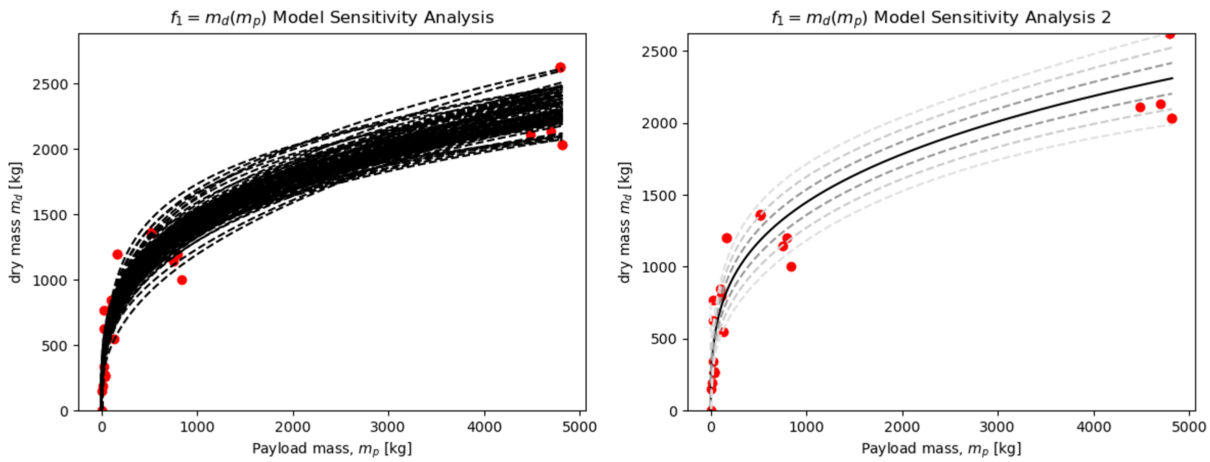

Figure 5: The result of cherry-pick fitting between the payload mass and the dry mass.

of this paper from one another.

$$m_{prpn} = m_{tanks} + m_{engines} + m_{plumbing} \tag{6}$$

The tanks on the spacecraft store fuel, oxidizer and pressurant. The mass of the tanks depends directly on their required volume, the storage pressure, the number of tanks the tank shape and the tank material. For the full tank sizing routine, see the project git-hub.[2]

The engines are sized using the following relation from Ramos in DYREQT in 2022.[26]

$$m_{engines} = \frac{T_r}{g_0} / ((6.098 \times 10^{-4}) \cdot T_r + 13.44) \cdot n \tag{7}$$

where T_r is the thrust requirement, g_0 is 9.81 and n is the number of engines. Lastly the plumbing is assumed to be 15% of the propulsion subsystem mass as in.[26]

The eleven sizing algorithms are compared to one another by comparing their predictions of the mass properties of landers in the database. The median value of the absolute percentage error is used as the figure of merit. The mean is not used because there are several outliers in the database which skew the mean error very high (above 100% in some cases.)

$$\tilde{x} = \text{median}\left(\{\frac{||\hat{y}_i - y_i||}{y_i}\}_{i=1}^{n}\right) \tag{8}$$

Where $\hat{y}_i$ is the prediction of the mass property of the *ith* lander, y_i is the actual value of the parameter, and n is the number of landers in the database.

2.3 Uncertainty Quantification

During the construction of these models to fit different parameters together, it may occur to the reader that the datasets used are quite small, and that the data is spread over a large time frame (1966 - 2025), there must be some amount of uncertainty associated with this data. Indeed, the uncertainty on the mass estimation models are of interest.

Cherry pick fitting was one method used to quantify this modelling uncertainty. A line of best fit is drawn between two variables. In figure 5, it is a logarithmic relationship of the following form:

$$y = \alpha \cdot ln(x) + \beta \tag{9}$$

The line is then redrawn but with one third of the data point randomly removed. This makes a slightly different line of best fit to the first. This process is repeated several hundred times. It yields a central line of best fit with additional knowledge of the distribution of the data above and below the line. This allows the designer to guess reasonable ranges for the dry mass in a Monte Carlo simulation to examine the stochastic properties of the sizing algorithm.

Table 4: Linear sizing rules based on data with outliers removed.

Subsystem	parameters	R^2
Structure	$m_{str} = 0.0747m_p$	0.8852
Avionics	$m_{avio} = 0.0774m_t$	0.833
Power	$m_{pow} = 0.018m_d + 311.68$	0.0914
Thermal	$m_{ther} = 0.0197m_t$	0.87
Propulsion	Component mass estimation	-
Other	$m_{oth} = 0.0316m_p$	0.9316

Table 5: Subsystem mass estimation parameters for multiple linear regression

Subsystem	a	b	c	d	e	intercept	R^2
Structure	-0.2100	0.0172	0.0746	-1.5635	7.9994	1,356	0.5431
Propulsion	0.6449	-0.0645	0.0100	1.4171	1.1680	-4,288	0.9547
Power	0.0818	0.0217	-0.0241	0.1914	-3.5158	897	0.8619
Avionics	0.1108	-0.0415	0.0406	-0.8425	1.0197	1,207	0.8728
Thermal	0.5412	-0.0130	-0.0714	0.2366	-6.7381	1,269	0.8976
Other	-0.1543	0.0910	-0.0276	0.4351	0.0745	-617	0.8947

3. Results and Discussion

There are three groups of results in this paper: first the sizing rules and the validation results for the major mass properties and subsystems are presented; Second the databases used to make the sizing rules are presented and the quality of the data discussed; Lastly, the three sizing algorithms are used to size the validation case.

3.1 Lunar Lander Sizing Rules

The parameters of all the sizing rules found in this paper are presented in the following tables along with the quality of their fits.

For the dry mass estimation based on the payload mass, the curve of best fit is:

$$m_d = 367.29 \cdot ln(m_p) - 904.17 \tag{10}$$

with an R^2 of 0.92.

For the propellant mass estimation, the adjusted Tsiolkovsky equation is used. $\Theta = 1.16$ is found to give the best fit to the data. This corresponds to a 16% propellant mass margin.

$$m_{prop} = 1.16 \cdot (m_p + m_d) \cdot (exp(\frac{\Delta V}{I_{sp} \cdot g_0}) - 1) \tag{11}$$

For the subsystems, all except the propulsion subsystem were sized by linear, or multiple linear regression using the subsystem mass database. The linear rules are shown in table 4.

Regarding the multiple linear regression, the reader is reminded of the following equation.

$$m_{ss} = a \cdot m_d + b \cdot m_{prop} + c \cdot m_p + d \cdot I_{sp} + e \cdot \Delta V + intercept \tag{12}$$

The corresponding values for the coefficients are shown in table 5.

The coefficient of determination of the fits for these MERs is less than in previous works on lunar landers such as Isaji 2020.[19] They have achieved R^2 of 0.9 or greater for all of subsystem MERs. There are two probable reasons for this: They used non-linear sizing rules found by curve fitting the data, and they used different input variables compared to this study: mission duration. Unfortunately the lunar lander database which they used to find these MERs is not public and so they are challenging to replicate.

Table 6: All successfully launched lunar lander missions in chronological order. Data also collected on ΔV, I_{sp}, and fuel type

Year	Organisation	Spacecraft name	Mission	mt	mp	md	mprop	source
1966	USSR	Luna 9	Ye-6M	1538	99.8	847	591.2	[14,28,48]
1966	NASA	Surveyor 1	Surveyor 1	995.2	33	262	700.2	[36,43]
1966	USSR	Luna 13	Ye-6M	1620	112	800	708	[13,14,27]
1967	NASA	Surveyor 5	Surveyor 5	1006	33	271	702	[38,43]
1967	NASA	Surveyor 6	Surveyor 6	1026	33	267	726	[39,43]
1967	NASA	Surveyor 3	Surveyor 3	1026	33	266	727	[37,43]
1968	NASA	Surveyor 7	Surveyor 7	1039	33	273	733	[40,43]
1969	NASA	LM-5 Eagle	Apollo 11	15103	4821	2034	8248	[1]
1970	USSR	Ye-8-5 (405)	Luna 16	5727	520	1360	3847	[13,14]
1969	NASA	LM-6 Intrepid	Apollo 12	15065	4819	2034	8212	[1]
1970	USSR	Ye-8 (203)	Luna 17 - Lunakhod 1	5660	756	1144	3760	[13,14]
1970	NASA	LM-7 Aquarius	Apollo 13	14916	4489	2109	8318	[1]
1971	NASA	LM-8 Antares	Apollo 14	15034	4700	2134	8200	[1]
1971	NASA	LM-10 Falcon	Apollo 15	16447	5295	2373	8780	[1]
1972	USSR	Ye-8-5 (408)	Luna 20	5750	520	1360	3870	[14,49]
1972	NASA	LM-11 Orion	Apollo 16	16447	5295	2373	8780	[1]
1972	NASA	LM-12 Challenger	Apollo 17	16447	5295	2373	8780	[1]
1973	USSR	Ye-8 (204)	Luna 21 - Lunakhod 2	5700	836	1000	3864	[13,14]
1976	USSR	Ye-8-5 (413)	Luna 24	5795	515	1365	3915	[14]
2013	CNSA	Chang'e 3	Chang'e 3	3780	170	1200	2410	[9]
2019	ISRO	Vikram 1	Chandrayaan-2	1471	27	626	818	[25]
2019	CNSA	Chang'e 4	Chang'e 4	3640	170	1200	2270	[9,42,50]
2019	SpaceIL	Beresheet	Beresheet	585	0.1	150	435	[41]
2020	CNSA	Chang' 5	Chang'e 5	3800	800	1200	1800	[24,44]
2022	JAXA	OMOTENASHI	OMOTENASHI	14.6	0.1	0.7	13.9	[21,22]
2023	Roscosmos	Luna 25	Luna 25	1750	30	770	950	[10,31,52]
2023	iSpace	Hakuto-R M1	Hakuto-R M1	1000	30	340	630	[35]
2023	ISRO	Vikram 2	Chandrayaan-3	1750	26	N/A	N/A	[30]
2024	JAXA	SLIM	SLIM	710	20	190	500	[20,46]
2024	Astrobotic	Peregrine	Peregrine Mission One	1283	90	743	450	[15]
2024	Intuitive Machines	Odysseus	IM-1	1908	130	494	1276	[4,16,33]
2024	CNSA	Chang'e 6	Chang'e 6	3800	800	1200	1800	[24,32,44]
2025	Firefly	Blueghost	Blueghost M1	1517	150	331	1036	[5–7]
2025	Intuitive Machines	Athena	Athena	2120	130	545	1445	[4,11,17]
2025	iSpace	Hakuto-R M2	Hakuto-R M2	1000	30	340	630	[29]

3.2 Lunar Lander Databases

The compiled databases used in this paper are shown here. Table 6 shows the major mass properties of every historical lunar lander launched towards the moon (excluding some early soviet and American models). Additional data collected includes the estimated ΔV, the I_{sp} and the number of stages.

The subsystem masses database relies entirely on one catalogue of conceptual lunar lander designs. It contains a mix of manned and unmanned systems with no models younger than 2005. Data on only unmanned lunar landers from the modern CLPS era (post 2018), is very difficult to find.

3.3 Lunar Lander Sizing Algorithms

The accuracy of the sizing algorithms was measured by their ability to predict the masses of the landers in the subsystem mass database using equation 8.

Eleven algorithms were tested and the three shown in table 8 are the best performing algorithms. They are chosen based on the total mass prediction error. The total mass is the most important parameter to predict for mission designers since this number will dictate the launch vehicle used to put the spacecraft in orbit. The errors are higher than those in

Table 7: Subsystem mass breakdown database. For two stage manned landers, only the masses present in the descent stage are counted. The total mass of the ascent stage is considered payload. All data from.[1] ND = No data

Lander	md	mp	mprop	dV	Isp	Structure	Propulsion	Power	Avionics	Thermal	Other
Apollo J-series	2373	5295	8780	2265	311	460	495	366	29	404	273
9508-HLR-1	1673.7	473.3	2891.6	1822	333	532.9	252.5	125.7	120.1	113.5	56
8801-EE-1 LOX-LH2	9823	25000	25252	2100	450	1681	4258	478	934	2017	455
8801-EE-1 N2O4/MMH	7899	25000	36399	2100	333	1955	2943	478	1062	1006	455
9205-FLO-1-CLS	9320	35894	44151	3100	450	1770	4969	385	307	420	1469
0503-CE&R-8	5858	16461	22502	2260	379	2822	1749	204	818	265	0
0609-LLPS-GRC	6503	229	4448	ND	450	1186.34	918.21	663	258.7	459.04	0
0610-LAT-1	10747	22074	50596	ND	450	453	2969	288	519	56	527
0507-ESAS-D	7542	28475	25486	1900	362	2015	2425	553	594	1072	883
0507-ESAS-F	7412	26375	23833	1900	351	1985	2347	553	594	1066	867
0507-ESAS-H	5970	22028	19152	1900	363	1877	1125	553	594	1049	772
0507-ESAS-J	5912	20956	18312	1900	363	1860	1095	553	594	1046	764

Table 8: Median absolute errors between the predictions of the sizing algorithms and the data. The best performing algorithms.

Mass property	N2O4 Algo 7 (%)	LOX/LH2 Algo 10 (%)	LOX/LCH4 Algo 11 (%)
m_p	0.00	0.00	0.00
m_t	3.63	13.29	5.51
m_d	28.67	31.02	34.855
m_{prop}	13.19	17.43	16.54
m_{str}	13.17	13.17	13.17
m_{prpn}	68.36	79.81	80.425
m_{avio}	64.04	53.04	66.59
m_{pow}	34.37	26.85	34.485
m_{ther}	17.56	21.49	24.79
m_{oth}	30.75	30.75	30.75

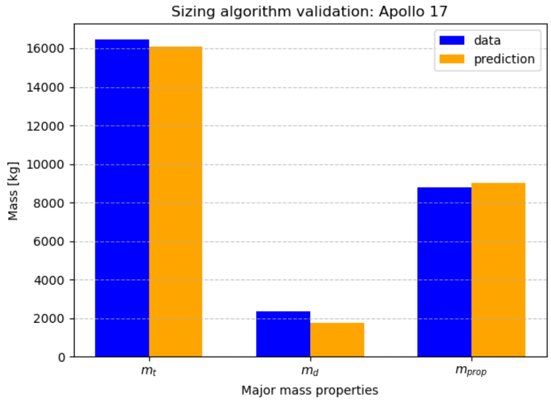

Figure 6: Validation of prediction of major mass properties: m_t, m_d and m_{prop}

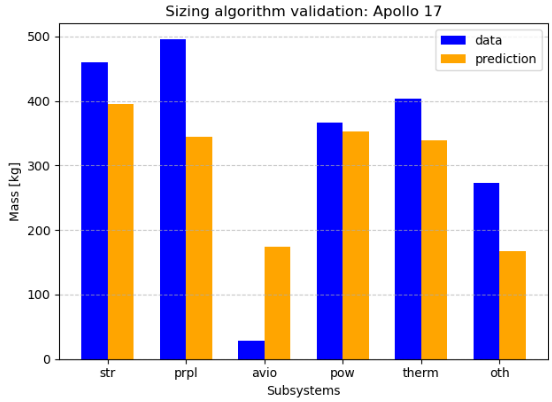

Figure 7: Validation of the prediction of subsystem mass properties.

the state of the art. There are three challenges which contribute to this: the first is that each algorithm was tested on all the members of the database, including landers which had different propellants. The second is that eleven out of twelve landers in the database are conceptual designs ranging from the 1980s to 2000s. Design philosophies, margin rules and technologies may have changed throughout that period and since none of them were flight tested, there may design errors present which would have been changed before production of the vehicles. Finally, the subsystem definitions of each lander are not consistent in the sources for the data which means some subsystem masses vary significantly from the mean.

In order to illustrate these challenges and the capabilities of the sizing algorithms, The N2O4 Algorithm 7 is validated against the Apollo 17 lunar module. Figure 6 and 7 shows a comparison between the prediction and the actual the mass properties of the LM.

4. Conclusion

This paper has presented three contributions in the area of lunar lander sizing: A fully cited public database of almost all flight models of lunar landers since 1966; validated sizing rules for estimating the final total mass, dry mass and propellant mass of a lunar lander based on its payload mass, ΔV and I_{sp}; and finally, three sizing algorithms for landers using three different propellants: N2O4/AZ, LOX/LH2 and LOX/LCH4.

Mass estimations as accurate as the state of the art cannot yet be achieved using these algorithms. More data, and investigation of non-linear sizing rules, and trial and error with other algorithms will yield better results.

The next steps for this work are quantifying the uncertainty on the subsystem MERs, propagating it through sizing algorithms to enable stochastic predictions of mass properties: eg. given a payload mass of X, there is a Y% probability of the total mass being below Z.

References

[1] John F Connolly. After lm: Nasa lunar lander concepts beyond apollo. Technical report, 2019.

[2] Conall de Paor. EUCASS_2025. `https://github.com/Conall98/EUCASS_2025`, 2025. GitHub repository.

[3] Conall de Paor, Jasmine Rimani, and Stephanie Lizy-Destrez. Low-cost uncrewed lunar landers: Mission design and preliminary feasibility assessment. In *Proceedings of the 10th European Conference for Aeronautics and Space Sciences (EUCASS)*, July 2023.

[4] Federal Communications Commission (FCC). Intuitive Machines-1 Orbital Debris Assessment Report (ODAR), 2024. Accessed: May 28, 2025.

[5] Firefly Aerospace. Blue Ghost Lunar Lander. `https://fireflyspace.com/blue-ghost/`, 2024. Accessed: March 29, 2025.

[6] Firefly Aerospace. FCC Licensing System Filing - Mass Waiver Request. https://licensing.fcc.gov/cgi-bin/ws.exe/prod/ib/forms/attachment_menu.hts?id_app_num=138525&acct=72303&id_form_num=12&filing_key=-462960, 2024. Accessed: May 28, 2025.

[7] Firefly Aerospace. BGM1 Component Graphic (Archived January 14, 2025), 2025. Accessed: May 28, 2025.

[8] Peter Fortescue, Graham Swinerd, and John Stark. *Spacecraft systems engineering*. John Wiley & Sons, 2011.

[9] Gunter Dirk Kremer. Chang'e 3. https://space.skyrocket.de/doc_sdat/change-3.htm, 2024. Accessed: 2025-05-27.

[10] Gunter's Space Page. Luna-Glob. https://space.skyrocket.de/doc_sdat/luna-glob.htm, 2023. Accessed: November 22, 2024.

[11] Gunter's Space Page. Nova-C (IM-2). https://space.skyrocket.de/doc_sdat/nova-c_im2.htm, 2024. Accessed: May 28, 2025.

[12] Joe Hamaker. Quickcost 5.0. https://www.iceaaonline.com/wp-content/uploads/2017/09/MM09_Presentation_QuickCost5.0_Hamaker.pdf, 2017. Presentation at the 2017 ICEAA Workshop, accessed: March 29, 2025.

[13] Brian Harvey. *Soviet and Russian lunar exploration*. Springer Science & Business Media, 2006.

[14] Wesley Theodore Huntress, Mikhail Iï¸ Aï¸¡kovlevich Marov, et al. *Soviet Robots in the Solar System: Mission Technologies and Discoveries*. Springer, 2011.

[15] Astrobotic Technology Inc. Payload user guide. https://web.archive.org/web/20190603165910/https:/www.astrobotic.com/payload-user-guide, 2019. Accessed: November 22, 2024.

[16] Intuitive Machines. Intuitive machines - advancing space exploration. https://www.intuitivemachines.com/, 2024. Accessed: October 9, 2024.

[17] Intuitive Machines. Nova-C Lunar Lander. https://www.intuitivemachines.com/nova-c, 2024. Accessed: May 28, 2025.

[18] Masafumi Isaji, Ian Maynard, and Bernd Chudoba. Surface access architecture modeling: Trend analysis and classification from a lander database. In *2018 AIAA SPACE and Astronautics Forum and Exposition*, page 5131, 2018.

[19] Masafumi Isaji, Ian Maynard, and Bernd Chudoba. A new sizing methodology for lunar surface access systems. In *AIAA Scitech 2020 Forum*, page 1775, 2020.

[20] Japan Aerospace Exploration Agency (JAXA). SLIM Media Kit. https://global.jaxa.jp/countdown/slim/SLIM-mediakit-EN_2308.pdf, 2023. Accessed: November 22, 2024.

[21] JAXA. Omotenashi lunar lander mission: Overview. https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=OMOTENASH, 2013. Accessed: March 8, 2023.

[22] JAXA. Omotenashi. https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=OMOTENASH, 2023. Accessed: March 8, 2023.

[23] Harry W Jones. Estimating the life cycle cost of space systems. In *International Conference On Environmental Systems*, number ARC-E-DAA-TN24212, 2015.

[24] Kaynouky. Chang'e 5. https://fr.wikipedia.org/wiki/Chang%27e_5, 2022.

[25] Chethan Kumar. Chandrayaan-2: All you need to know about india's 2nd moon mission. Available online at: https://timesofindia.indiatimes.com/india/chandrayaan-2-all-you-need-to-know-about-indias-2nd-moon-mission/articleshow/70207662.cms, 2019. The Times of India, Accessed: 2022-02-28.

[26] Eugina Mendez Ramos, Bradford E Robertson, Manuel J Diaz, and Dimitri N Mavris. Development of a lunar lander modeling and simulation capability. In *AIAA Scitech 2022 Forum*, page 0380, 2022.

[27] NASA Space Science Data Coordinated Archive. Luna 13. https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1966-116A, 1966. Accessed: 2025-05-28.

[28] NASA Space Science Data Coordinated Archive. Luna 9. https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1966-006A, 1966. Accessed: 2025-05-28.

[29] NASA Space Science Data Coordinated Archive (NSSDC). Spacecraft Details: Crew Dragon Resilience. https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=RESILIENC, 2020. Accessed: May 28, 2025.

[30] NASA Space Science Data Coordinated Archive (NSSDC). Spacecraft Details: Chandrayaan-3. https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=2023-098A, 2023. Accessed: May 28, 2025.

[31] NASA Space Science Data Coordinated Archive (NSSDC). Spacecraft Details: Luna 25. https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=2023-118A, 2023. Accessed: May 28, 2025.

[32] NASA Space Science Data Coordinated Archive (NSSDC). Future Chinese Lunar Missions (CNSA). https://nssdc.gsfc.nasa.gov/planetary/lunar/cnsa_moon_future.html, 2024. Accessed: May 28, 2025.

[33] NASA Space Science Data Coordinated Archive (NSSDC). IM-1 NOVA Mission Information. {https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=IM-1-NOVA}, note = Accessed: November 22, 2024,, 2024.

[34] NASA Space Science Data Coordinated Archive (NSSDC). Lunar Exploration Timeline. https://nssdc.gsfc.nasa.gov/planetary/lunar/lunartimeline.html, 2024. Accessed: May 28, 2025.

[35] National Space Science Data Center (NSSDC). Hakuto-RM1. https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=2022-168A, 2022. [Online; accessed 8-March-2023].

[36] NSSDC. Surveyor 1. https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1966-045A. Accessed 8-February-2023.

[37] NSSDC. Surveyor 3. https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1967-035A. accessed: 28/02/2022.

[38] NSSDC. Surveyor 5. https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1967-084A. Accessed: 28/02/2022.

[39] NSSDC. Surveyor 6. https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1967-112A. Accessed: 28/02/2022.

[40] NSSDC. Surveyor 7. https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1968-001A. Accessed: 28/02/2022.

[41] NSSDC. Beresheet. https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=2019-009B, 2019. Accessed: 28/02/2022.

[42] NSSDC. Chang'e 4. https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=2018-103A, 2019. Accessed: 17/6/2022.

[43] United States. Surveyor Program Office. *Surveyor Program Results*, volume SP-184 of *NASA Special Publication*. Scientific and Technical Information Division, National Aeronautics and Space Administration, Washington, D.C., 1969. Accessed: 2025-05-28.

[44] Spaceflight101. Chang'e 5. https://web.archive.org/web/20240714023515/https://spaceflight101.com/change/change-5/, 2020.

[45] Douglas Stanley, Stephen Cook, John Connolly, and Jeffrey Hanley. Exploration systems architecture study: Overview of architecture and mission operations approach. In *SpaceOps 2006 Conference, 2006*, page 5935, 2006.

[46] The Planetary Society. SLIM: Japan's Precision Lunar Lander. https://www.planetary.org/space-missions/slim-japans-precision-lunar-lander, 2024. Accessed: November 22, 2024.

[47] Unison Global. Unison Global - Homepage. https://www.unisonglobal.com/, 2024. Accessed: March 29, 2025.

[48] Mark Wade. Luna e-6. http://www.astronautix.com/l/lunae-6.html. Online; accessed 8-March-2023.

[49] Mark Wade. Luna ye 8-5. http://www.astronautix.com/l/lunaye-8-5.html. Accessed: March 8, 2023.

[50] Qiong Wang and Jizhong Liu. A chang'e-4 mission concept and vision of future chinese lunar exploration activities. *Acta Astronautica*, 127:678–683, 2016.

[51] James Richard Wertz, Wiley J Larson, Douglas Kirkpatrick, and Donna Klungle. *Space mission analysis and design*, volume 8. Springer, 1999.

[52] Wikipedia contributors. Luna 25 – WikipÃ©dia. https://fr.wikipedia.org/wiki/Luna_25, 2024. Accessed: May 28, 2025.